

Math 243: Practice Midterm #1

Instructions: you must show your work to receive credit. You have 50 minutes. Calculators are not allowed.

Problem 1 (Differentiation of curves). Consider the curve

$$\begin{cases} x(t) = 5t^2 - 6t + 4 \\ y(t) = t^2 + 6t - 1 \end{cases}$$

- (a) Find an equation of the tangent line at $t = 0$.
- (b) For what values of t is this curve concave down? For what values is it concave up?

Problem 2 (Arc length). Compute the arc length of the parametrized curve:

$$\begin{cases} x(t) = \frac{4}{3}t^3 + t - 2 \\ y(t) = \frac{1}{27}(9t^2 + 2)^{\frac{3}{2}} \end{cases}$$

Problem 3 (Polar coordinates). Consider the polar graph

$$\begin{cases} r = 6 \sin(\theta) \\ 0 \leq \theta \leq \pi \end{cases}$$

Convert this equation to Cartesian form and plot it.

Problem 4 (Projections). Let $\vec{u} = \langle 1, -1, 1 \rangle$ and $\vec{v} = \langle -2, -2, -2 \rangle$.

- (a) Find the projection of $\vec{u}$ along $\vec{v}$.
- (b) What is the cosine of the angle between $\vec{u}$ and $\vec{v}$?
- (c) Find both unit vectors orthogonal to $\vec{u}$ and $\vec{v}$.

Problem 5 (Planes). Consider the points $A = (2, 5, -1)$, $B = (-2, 1, 3)$ and $C = (7, 1, 1)$.

- (a) Find an equation of the line passing through points A and B .
- (b) Find an equation the plane passing through points A , B , and C .

Problem 6 (Sphere).

- (a) State the Cartesian formula for a sphere.
- (b) Show that the equation $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$ represents a sphere.
- (c) What is the center and radius of the sphere?

Math 243: Practice Midterm #2

Instructions: you must show your work to receive credit. You have 50 minutes. Calculators are not allowed.

Problem 1 (Vectors).

- (a) Compute the distance between the following two points $(5, 8, 15)$ and $(1, 4, 8)$.
- (b) Are the vectors $\langle 2, 4, 7 \rangle$ and $\langle 4, 8, 14 \rangle$ orthogonal, parallel, or neither? Justify your answer.
- (c) Are the vectors $\langle 3, -1, 3 \rangle$ and $\langle 5, 9, -2 \rangle$ orthogonal, parallel, or neither? Justify your answer.

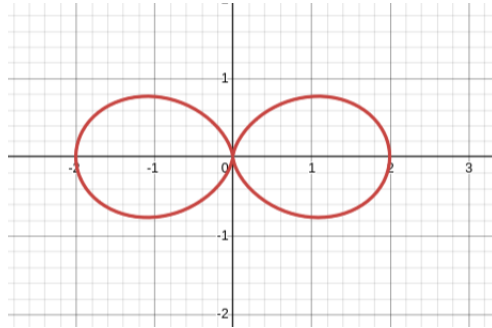
Problem 2 (Arc length). Find the arc length of the parametric curve

$$\begin{cases} x(t) = \frac{1}{2}t^2 - t - 1 \\ y(t) = \frac{4}{3}t^{\frac{3}{2}} \end{cases}$$

on the interval $1 \leq t \leq 2$

Problem 3 (Polar stuff). Find the area enclosed by the curve

$$\begin{cases} r = 1 + \cos(2\theta) \\ 0 \leq \theta \leq \pi \end{cases}$$



Problem 4 (Projections). Consider the point $W = (1, 0, 3)$ and the plane given by the equation $5x - y - z = 3$. Find the shortest distance from W to the plane. Include a relevant sketch in your answer.

Problem 5. Let C be the curve defined by the parametric equations

$$\begin{cases} x(t) = t^2 - t + 2 \\ y(t) = t - \frac{1}{3}t^3, \end{cases}$$

where $-10 \leq t \leq 10$.

- (a) Find the equation of the tangent line to C at $t = 2$.
- (b) For what values of t is the tangent line horizontal? For what values is it vertical?
- (c) Compute $\frac{d^2y}{dx^2}$.
- (d) Is C concave up or concave down when $t = 1$? You must justify your answer to receive credit.

Problem 6 (Planes and cross product). The planes given by the equations

$$4x - y + 3z = 1 \quad \text{and} \quad 2x - 2y + z = 1$$

intersect to form a line L .

- (a) Find a vector in the direction of the line L .
- (b) Give an example of a point on the line L .
- (c) Find an equation of the line L (either a vector equation or a parametric equation is fine).